

KUNAL PATEL

SENIOR AI/ML ENGINEER & AUTOMATION SYSTEMS ARCHITECT

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with an MSc in Artificial Intelligence & Machine Learning and proven expertise in designing, deploying, and scaling production AI applications. Specialized in multi-agent LLM orchestration (LangGraph, Groq LLaMA 3.3 70B, OpenAI, Gemini), Retrieval-Augmented Generation (RAG) vector search pipelines, custom YOLOv8 computer vision models, and automated FastAPI backends. Proven track record of architecting 9+ production AI platforms and engineering enterprise automation pipelines with 40%+ operational efficiency gains.

CORE COMPETENCIES & TECHNICAL SKILLS

- **AI & Multi-Agent Systems:** LangGraph (Multi-Agent Workflows), LangChain, Groq LLaMA 3.3 70B, OpenAI API, Google Gemini API, RAG Architecture, Vector Embeddings (ChromaDB), Prompt Engineering, BYOK Key Gateway.
- **Machine Learning & Computer Vision:** PyTorch, YOLOv8 (Real-Time Defect & Occupancy Detection), OpenCV, Image Segmentation, Bounding Box Annotation, Model Fine-Tuning, Performance Evaluation.
- **Backend & Cloud Architecture:** Python, FastAPI, TypeScript, Next.js, React, Tailwind CSS, REST APIs, JSON API Schemas, SQL, Docker, Postman, Git/GitHub, Render, Vercel.
- **Workflow Automation & Integration:** n8n, Make, Zapier, Web Scraping Pipelines, Microservice Webhooks, Automated ETL & Data Extraction Pipelines.

PROFESSIONAL EXPERIENCE

AI-ML Engineer

May 2026 – Present

Capermint Technology | Ahmedabad, Gujarat

- Architected real-time AI/ML game engines and dynamic NPC interaction behaviors, reducing model inference latency by 35% across mobile and web platforms.
- Engineered telemetry-driven data pipelines and automated user engagement scoring systems, increasing D1 user retention by 22%.
- Integrated high-throughput REST API microservices in FastAPI, seamlessly connecting ML inference endpoints with front-end game loops.

AI Engineer

Dec 2025 – May 2026

Elite Workforces Services | Ahmedabad, Gujarat

- Spearheaded end-to-end enterprise workflow automation using Python, REST APIs, and n8n, automating 40%+ of manual processes and saving 120+ engineering hours/month.
- Integrated multi-provider LLM fallback gateways (OpenAI, Groq LLaMA) with automated failover handling and real-time execution analytics.
- Built custom performance analytics dashboards, translating raw workflow execution logs into actionable operational insights for client stakeholders.

AI Engineer

May 2025 – Nov 2025

One Percent Media | Ahmedabad, Gujarat

- Designed high-throughput AI content automation pipelines using custom Python scripts and n8n webhooks, handling 5,000+ daily operational requests.
- Engineered automated OCR & NLP data extraction tools, reducing document processing turnaround time from hours to seconds with 98% accuracy.

AI Automation Engineer

Dec 2024 – May 2025

Sevensseed Technology | Ahmedabad, Gujarat

- Engineered JSON API-driven automation engines compatible with n8n, Make, and Zapier, orchestrating micro-app workflows across 7+ SaaS platforms.

- Standardized workflow error-handling protocols and API webhooks, eliminating pipeline drop-offs and maintaining 99.8% background task reliability.

FEATURED PRODUCTION AI PROJECTS

Sevensed AI Multi-Agent Startup Ecosystem

[Live Project ↗](#)

TECH STACK: LangGraph • Groq LLaMA 3.3 70B • ChromaDB RAG • FastAPI

- Architected a multi-agent SaaS platform orchestrating 9 autonomous AI startups (Comonk, Sevenforce, Breakdown Factor, Rakshak AI, AVP Emart, AVPU, Decode Forest, Trust) using LangGraph, Groq LLaMA 3.3 70B, ChromaDB vector RAG, and FastAPI.
- Implemented a self-service Bring-Your-Own-Key (BYOK) key manager in Next.js, enabling zero-cost client workloads across Gemini, OpenAI, and Groq.

Comonk AI - Enterprise Career Intelligence Platform

[Live Project ↗](#)

TECH STACK: RAG Matching • ATS Resume Optimizer • Mock Interview Copilot

- Developed a 32-panel AI career copilot featuring RAG job matching, ATS resume optimizer, voice/video mock interview scoring with PDF reports, and automated application tracking.

Sevenforce - Autonomous AI Workforce & Sales CRM

[Live Project ↗](#)

TECH STACK: 7-Agent Workforce Dock • Lead Scoring • Bot Automation

- Built a 7-agent AI workforce dock (Maya, Sales CRM, automated lead scoring, email automation, meeting bot) for enterprise workflow automation.

Breakdown Factor - YOLOv8 Structural Defect Scanner

[Live Project ↗](#)

TECH STACK: YOLOv8 best.pt • PyTorch • OpenCV • IS-456 BOQ Estimation

- Fine-tuned a custom YOLOv8 PyTorch model (best.pt) for real-time 10+ category structural defect detection (cracks, pipe leaks, tile damage) and BOQ material cost estimation.

Rakshak AI - Citizen Assistant, Police Copilot & Vision Security Suite

[Live Project ↗](#)

TECH STACK: Automatic FIR (BNS/IPC) • Multilingual Chatbot • PPE Mask Scanner • YOLO

- Built a 5-in-1 AI platform combining automatic FIR generation with Bharatiya Nyaya Sanhita (BNS/IPC) legal code recommendations, multilingual AI chatbot, cybercrime scam analyzer, and emergency SOS dispatch.
- Integrated Computer Vision workstations for safety mask PPE compliance scanning, sub-second facial attendance verification, and YOLO chair/occupancy detection.

EDUCATION & CERTIFICATIONS

MSc in Artificial Intelligence & Machine Learning

Aug 2024 – Apr 2026

Sardar Patel University | Anand, Gujarat

- Specialized coursework in AI systems, deep learning architectures, predictive modeling, computer vision, and autonomous agent systems.

Bachelor of Computer Applications (BCA)

Jun 2020 – Apr 2023

Dharmsinh Desai University | Nadiad, Gujarat

- Foundational training in software engineering, database management systems, data structures, algorithms, and web development.